

Retrieval And Subgroups

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Problem 1. *Benjamini–Hochberg at $q = 0.05$ with $m = 5$ tests. The sorted p -values are $p_{(1)}, \dots, p_{(5)} = 0.004, 0.030, 0.045, 0.047, 0.060$. The bar for rank i is $\frac{i}{m}q$. Find the largest i with $p_{(i)} \leq \frac{i}{m}q$ and reject ranks 1 to i , where “rejected” means the null is rejected and the contrast is called significant. Which ranks does BH call significant?*

Solution. Only the lowest rank is significant.

rank i	$p_{(i)}$	bar $\frac{i}{5}(0.05)$	$p_{(i)} \leq \text{bar?}$
1	0.004	0.01	yes
2	0.030	0.02	no
3	0.045	0.03	no
4	0.047	0.04	no
5	0.060	0.05	no

The largest passing rank is 1, so BH rejects rank 1 alone.